

Arvind Shastri

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Product-focused software engineer with founder experience building and shipping production software from 0-to-1.

WORK EXPERIENCE

Trac

Jun 2025 – Present

Founding Software Engineer

Hamilton, ON

- Built and scaled a **0-to-1** real-time shuttle tracking application to **4000+ active users**, securing a **B2B** partnership with Attridge Transportation and deploying into production at McMaster University 
- Developed a **recency-weighted multiple linear regression** pipeline in **Python** and **scikit-learn**, ingesting live bus telemetry to predict shuttle ETAs **within 30 seconds** of actual arrival
- Engineered an **event-driven** architecture for live shuttle bus tracking via **Firebase** to broadcast telemetry data to 100+ concurrent users at **3-second intervals**
- Implemented live vehicle tracking in **Flutter** with **Mapbox**, optimizing multi-layer map annotations and stateful tap-to-follow camera logic to achieve **sub-500ms rendering latency**
- Led technical discovery across 15 stakeholder interviews to identify pain points, designing 10+ rapid prototypes in **Figma Make** that achieved **95%+ usability satisfaction**

Ericsson

May 2023 – Aug 2024

Software Developer Co-op

Ottawa, ON

- Delivered client-facing features for an enterprise network design platform, integrating front-end components with APIs using **React**, **Redux**, and **TypeScript** to resolve **50+** issues across 4 product releases
- Optimized 2D CAD data processing by using **Python OpenCV** and **Tkinter** with C++ data reduction algorithms, cutting image generation time by **10%**
- Accelerated CI/CD pipeline efficiency through **AI-assisted** coding workflows to optimize front-end UI automations in **Nightwatch.js**, reducing **Jenkins E2E** test suite runtime by **15%**
- Led a team of 3 to scope the migration of a **MATLAB app** to React/TypeScript by gathering and translating stakeholder requirements into 6 high-fidelity **Figma** prototypes, delivering the PoC **2 weeks early**

CGI

May 2022 – Aug 2022

Software Developer Co-op

Remote

- Refactored **Java** backend logic within a **JSP** calendar application to automate event workflows, eliminating manual approval bottlenecks and reducing scheduling turnaround time by **40%**
- Implemented a multi-attribute filtering system in **Java** and **JSP** to query **1000+** calendar records by multiple attributes, reducing user search time by **60%**
- Built responsive feedback components in **HTML/CSS**, resolving 5 high-priority customer-reported issues and surfacing clear system feedback on previously ambiguous UI actions
- Optimized **REST API** calls by parallelizing sequential fetches, reducing page load latency by **50%**

PROJECTS

Court Report – AI-Powered NBA Daily Digest | *Python, LangChain, Chroma, Claude API, FastAPI, React* Jun 2026

- Built a **RAG pipeline** using **LangChain** and **Chroma** to semantically store and retrieve historical NBA game context, grounding **Claude API**-generated digests in real performance data
- Developed a **React** frontend and **FastAPI** backend featuring a **RAG-powered chatbot** with type-based vector retrieval, multi-turn conversation history, and pre-retrieval query enrichment
- Applied **prompt engineering** with strict output constraints to eliminate statistic hallucinations, validated through an **LLM-as-Judge** evaluation suite across 16 diverse test cases averaging **94%**

SKILLS

Languages: Python, Dart, TypeScript, HTML, CSS, JavaScript, Java, SQL

Tools and Frameworks: Flutter, React, Redux, Firebase, Mapbox, FastAPI, Git, GitHub, CI/CD

AI/ML: Claude Code, Claude API, LangChain, Chroma, RAG, prompt engineering, LLM evaluation

Product and Design: Figma, product scoping, wireframing, high-fidelity prototyping, usability testing

EDUCATION

McMaster University

Jun 2025

Bachelor of Engineering, Software Engineering, Co-op 

Hamilton, ON